

BLUEPRINT TECNICO MESTRE

bundle.social

Analise tecnica completa para alternativa superior

114 endpoints | 15 plataformas | OpenAPI 3.0 real
Limites de midia | Resolucoes | Fluxo de postagem | Paralelismo
Rotas | Tempos | Mapeamento | Validacao | Erros

Fontes: [api.bundle.social/swagger-json](#) + [docs.bundle.social](#) + scan VulnStrike

Data: 25/08/2026

1. VISAO GERAL E URLs OFICIAIS

URLs Oficiais

Recurso	URL
Site	https://bundle.social
API prod	https://api.bundle.social
API dev	http://localhost:3001
Docs	https://docs.bundle.social
Docs MD	https://info.bundle.social
OpenAPI JSON	https://api.bundle.social/swagger-json
OpenAPI YAML	https://api.bundle.social/swagger-yaml
SDK	github.com/bundleglobal/bundlesocial-node
Status	bundlesocial.betteruptime.com
Contato	contact@bundle.social

Dados OpenAPI

Campo	Valor
Titulo	bundle.social API
Versao	1.0.0
Spec	OpenAPI 3.0
Endpoints	114
Plataformas	15
Auth	API Key (x-api-key header)
Tags	app, organization, team, socialAccount, upload, post, postImport, analytics, comment, misc, postCSV

15 Plataformas

TIKTOK | YOUTUBE | INSTAGRAM | FACEBOOK | TWITTER | THREADS | LINKEDIN | PINTEREST | REDDIT | MASTODON | DISCORD | SLACK | BLUESKY | GOOGLE_BUSINESS | SNAPCHAT

1. VISAO GERAL E URLs OFICIAIS

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2. AUTENTICACAO

2.1 API Key (cliente -> bundle.social)

Header unico: x-api-key: pk_live_XXX. 401=sem chave, 403=chave invalida. Chaves sao ORG-level (acesso a todos os teams). Prefixo pk_live_ (padrao Stripe-like). Sem OAuth2, sem JWT.

2.2 OAuth (bundle.social -> plataformas)

OAuth feito ENTRE bundle.social e cada plataforma. Cliente nao ve tokens.

Fluxo Hosted (recomendado)

1. POST /social-account/create-portal-link { teamId, redirectUrl, socialAccountTypes[] }
2. bundle.social retorna URL portal
3. Usuario abre portal - gerencia OAuth UI + canal + idioma
4. Redirecionado de volta para redirectUrl
5. Webhook social-account.created dispara

Fluxo Custom UI

1. POST /social-account/connect { type, teamId, redirectUrl, serverUrl?, disableAutoLogin?, tiktokForceLogin? } -> { url }
2. Redireciona usuario para URL OAuth da plataforma
3. Plataforma faz callback para bundle.social
4. POST /social-account/set-channel { teamId, type, channelId, channelData }
5. POST /social-account/refresh-channels
6. Webhook social-account.created

2.3 Token Management

Armazenado por conta: accessToken (ENCRIPADO), refreshToken (ENCRIPADO), secret, expiresAt.

REGRAS:

- NUNCA aparecem em webhooks ou respostas publicas
- Renovacao automatica antes de expirar
- Deteccao desconexao remota a cada 6h (Meta: FB, IG, Threads)
- 3 retries com 10min backoff se validacao falhar
- Apos 3 falhas: agenda delecao + webhook social-account.updated
- Grace period 6h antes de deletar
- POST /social-account/connection-check valida manualmente

2.4 Webhook Signature

Header: x-signature: HMAC-SHA256(payload, signing_secret)

Verificacao Node.js:

```
const exp = crypto.createHmac("sha256", secret)
    .update(JSON.stringify(payload)).digest("hex");
crypto.timingSafeEqual(Buffer.from(sig), Buffer.from(exp))
```

3. HIERARQUIA MULTI-TENANT

Estrutura

Organization (cobranca)

- API Keys (compartilhadas entre teams)
- Webhooks (compartilhados entre teams)
- Subscription (FREE/PRO/BUSINESS/CUSTOM)
- Team A ("Marketing")
 - Social Accounts (IG, TikTok, ...)
 - Posts, Uploads, Comments
- Team B ("Cliente: Acme")
 - Social Accounts proprias
 - Posts, Uploads, Comments

REGRA CRITICA:

- Rate limits API: por ORG (API key)
- Daily post limits: por CONTA REAL (platform + account_id)
- Monthly caps: por ORGANIZATION
- Para SaaS: 1 TEAM por cliente

Planos

Plano	Preco	Posts/mes	Comments	Uploads	Contas
FREE	\$0	20	50	200	3
PRO	\$100/mes	10.000	5.000	100.000	Ilimitado
BUSINESS	\$400/mes	100.000	50.000	1.000.000	Ilimitado
CUSTOM	Sob consulta	Custom	Custom	Custom	Custom

Contagem mensal: por data de CRIACAO no mes UTC. Rascunhos nao contam. Falhas NAO devolvem cota. Reset: dia 1 UTC.

4. FLUXO COMPLETO DE POSTAGEM (inicio ao fim)

Fase 1: Request do Cliente

```
POST /api/v1/post/
Headers: x-api-key: pk_live_xxx, Content-Type: application/json
Body:
{
  "teamId": "team_123",
  "title": "Lancamento X",
  "postDate": "2026-09-01T15:00:00.000Z",
  "status": "SCHEDULED",
  "socialAccountTypes": ["INSTAGRAM", "TIKTOK", "TWITTER", "LINKEDIN"],
  "data": {
    "INSTAGRAM": { "type": "REEL", "text": "BTS!", "uploadIds": ["upl_1"], "shareToFeed": true },
    "TIKTOK": { "type": "VIDEO", "text": "Launch!", "uploadIds": ["upl_1"], "privacy": "PUBLIC_TO_EVERYONE" },
    "TWITTER": { "text": "Shipping day!", "uploadIds": ["upl_2"] },
    "LINKEDIN": { "text": "We shipped X.", "privacy": "PUBLIC" }
  },
  "firstComment": { "INSTAGRAM": "#launch", "TIKTOK": "Link in bio!" }
}

Obrigatorios: teamId, title, postDate, status, socialAccountTypes, data
```

Fase 2: Validacao Inicial (sincrona, <100ms)

1. Validar API key (hash lookup no DB)
2. Validar rate limit (3 camadas Redis: 1s/100, 10s/500, 60s/2000)
3. Validar monthly cap (posts criados este mes < limite do plano)
4. Validar daily limit por plataforma (por conta real, nao por conexao)
5. Validar schema do body (Zod/Joi)
6. Validar uploadIds existem e pertencem ao team
7. Validar regras por plataforma (aspect ratio, tamanho, duracao)
8. Se postDate no passado -> 400
9. Salvar no DB como SCHEDULED
10. Retornar 201 com { id: "post_abc123", status: "SCHEDULED" }

Fase 3: Agendamento (BullMQ/Queue)

1. Scheduler pega posts com status=SCHEDULED e postDate <= now
2. Cria job na fila "publish"
3. Job contem: postId, teamId, socialAccountTypes[], data
4. Worker pega job (concorrencia tipicamente 50)

Fase 4: Publisher Worker (PARALELO por plataforma)

CADA plataforma publicada em paralelo (Promise.allSettled). Falha em uma NAO afeta outras.

```
async function publishPost(post) {
  const results = {};
  const platforms = post.socialAccountTypes;

  // PARALELO em todas as plataformas
  await Promise.allSettled(platforms.map(async (platform) => {
    const adapter = adapters[platform];
    const account = await getSocialAccount(post.teamId, platform);
    const platformData = post.data[platform];

    try {
      await adapter.ensureValidToken(account);
      const mediaUrls = await resolveUploadIds(platformData.uploadIds);
      const result = await adapter.publish(platformData, account, mediaUrls);
    } catch (error) {
      // tratamento de erro
    }
  }));
}
```

```
    results[platform] = { status: "POSTED", externalId: result.id, permalink: result.permalink };
  } catch (err) {
    results[platform] = { status: "ERROR", error: normalizeError(err, platform) };
  }
}));

const allPosted = Object.values(results).every(r => r.status === "POSTED");
post.status = allPosted ? "POSTED" : "ERROR";
post.externalData = extractExternalIds(results);
post.errorsVerbose = extractErrors(results);
post.postedDate = new Date();
await post.save();

await triggerWebhook("post.published", post);
if (post.firstComment) await scheduleFirstComment(post);
}
```

Fase 5: Pos-Publicacao

1. Webhook post.published disparado (status POSTED ou ERROR)
2. firstComment: criado apos publicacao bem-sucedida (cada plataforma tem maxLength)
3. Status PROCESSING: video grande processando na plataforma - re-check periodicamente
4. posts:retry <id> para re-tentar apos falha transient
5. Analytics: primeiro fetch em ate 24h depois da publicacao

5. PARALELISMO E PUBLISHER PIPELINE

5.1 Arquitetura

- CADA plataforma em socialAccountTypes[] roda em paralelo (Promise.allSettled).
- Post para 15 plataformas = 15 chamadas simultaneas
 - Tempo total = max(plataforma_mais_lenta)
 - Falha em Instagram NAO bloqueia TikTok
 - Cada plataforma tem proprio timeout, retry e modo de falha

5.2 Tempos Estimados por Plataforma

Plataforma	Tempo tipico	Timeout	Observacao
Instagram	2-5s	30s	Meta Graph API, video pode processar por min
Facebook	2-5s	30s	Meta Graph API, similar IG
TikTok	5-30s	60s	Pode entrar REVIEW (min a horas)
YouTube	10s-5min	300s	Upload video grande e lento
Twitter/X	1-3s	15s	API rapida
LinkedIn	2-5s	30s	API rapida
Threads	2-5s	30s	Meta infra
Pinterest	3-8s	30s	Processa imagem/video
Reddit	2-5s	20s	API rapida
Mastodon	1-3s	15s	Depende da instancia
Bluesky	1-3s	15s	AT Protocol, rapido
Discord	1-2s	10s	Webhook, instantaneo
Slack	1-2s	10s	Webhook, instantaneo
Google Business	3-8s	30s	Google API
Snapchat	5-15s	60s	Upload video

5.3 Status do Post

Status	Significado	Quando
DRAFT	Rascunho, nao agendado	Cliente salva sem postDate
SCHEDULED	Agendado	Cliente define postDate + status=SCHEDULED
PROCESSING	Publicando/processando	Worker pegou job, enviou para plataforma
POSTED	Publicado com sucesso	Plataforma confirmou
ERROR	Falhou permanentemente	Erro non-transient ou retries falharam
REVIEW	Plataforma revisando	TikTok colocou em revisao

5.4 Retry Logic

1. Erro TRANSIENT (rate limit, timeout, 5xx): retry automatico
 - Backoff exponencial: 30s, 90s, 270s (3 tentativas)
 - Max 3 retries automaticos
2. Erro NON-TRANSIENT (auth, validacao): sem retry
 - Cliente deve corrigir (reconectar, mudar midia)
 - POST /post/{id}/retry para re-tentar apos correcao
3. retryCount incrementado a cada tentativa
4. Apos 3 retries transient falhando: status=ERROR

6. SISTEMA DE UPLOAD (3 metodos + from-url)

6.1 Resumo

Metodo	Best for	Ceiling	Retryable	Endpoint
Simple	Imagens, small clips	90 MB	Nao	POST /upload/
Direct	Files moderados	5 GiB	Nao (recomeca)	POST /upload/init + finalize
Multipart	Videos grandes	5 GB default	Sim (por chunk)	POST /upload/multipart/*
From URL	Midia em CDN	1 GB	Nao	POST /upload/from-url

6.2 Simple Upload (multipart/form-data)

```
POST /api/v1/upload/
Content-Type: multipart/form-data
Fields: teamId (opcional), file (binary)
Retorna: { id: "upload_abc123", fileName, mimeType, size }
Unico endpoint com multipart/form-data.
```

6.3 Direct Upload (presigned URL, ate 5 GiB)

```
PASSO 1: POST /api/v1/upload/init
Body: { fileName, mimeType, fileSize?, teamId? }
Retorna: { url, path }
URL presigned expira em 30 MINUTOS
```

```
PASSO 2: PUT <url> --upload-file video.mp4
Enviar bytes crus (NAO JSON, NAO multipart)
Se falhar aos 99%: recomeca do zero
```

```
PASSO 3: POST /api/v1/upload/finalize
Body: { path, teamId? }
Retorna: { id: "upload_abc123", ... }
```

6.4 Multipart Upload (chunks 64 MiB, retryable)

```
PASSO 1: POST /api/v1/upload/multipart/init
Body: { fileName, mimeType, fileSize, teamId? }
fileSize REQUIRED (rejeita oversized aqui)
Retorna: { uploadId, path, partSize: 67108864, parts: [{partNumber, url}] }
URLs presigned expiram em 6 HORAS
MAX 10.000 parts
```

```
PASSO 2: PUT <part1.url> --upload-file chunk1
Cada chunk EXATAMENTE partSize bytes (exceto ultimo)
LER header ETag de cada PUT (necessario para complete)
Cada chunk independente - retry individual
```

```
PASSO 3 (se URL expirou): POST /api/v1/upload/multipart/sign
Body: { path, uploadId, partNumbers: [3, 7] }
Re-assina apenas partes que falharam
Partes ja enviadas MANTEM ETags
```

```
PASSO 4: POST /api/v1/upload/multipart/complete
Body: { path, uploadId, parts: [{partNumber, etag}], teamId? }
Retorna: { id: "upload_abc123", ... }
```

```
PASSO 5 (desistir): POST /api/v1/upload/multipart/abort
Body: { path, uploadId }
Partes nao finalizadas ocupam storage
Auto-abort apos 7 dias
```

6.5 Upload from URL

POST /api/v1/upload/from-url

Body: { url, teamId? }

url: URL publica HTTP(S) - bundle.social faz fetch server-side

Max size: 1 GB | Download timeout: 60 segundos

6.6 Storage e Cleanup

Storage: S3-compativel (R2 ou S3)

Uploads deletados: removidos apos 7 dias

Uploads nao usados: soft-delete 90 dias + 7 dias purge

Multipart nao completado: auto-abort 7 dias

Midia processada server-side: transcoding, validacao, redimensionamento

Posts podem ficar PROCESSING se video grande processando na plataforma

7. LIMITES DE MIDIA POR PLATAFORMA

7.1 Instagram - Posts (Carousel)

Constraint	Valor
Files	1-10 (imagens e/ou videos, misturados)
Image max size	8 MB
Image max width	1920px
Image aspect ratio	4:5 a 1.91:1
Video max width	1920px
Video max bitrate	45 Mbps
Video duration	3s - 15 min
Video aspect ratio	0.01:1 a 10:1

Reels

Constraint	Valor
Files	1 video (required)
Max width	1920px
Max bitrate	45 Mbps
Duration	3s - 15 min
Aspect ratio	0.01:1 a 10:1 (9:16 recomendado)

Stories

Constraint	Valor
Files	1 (image or video)
Image max size	8 MB
Image aspect ratio	0.01:1 a 10:6
Video max size	100 MB
Video max width	1920px
Video max bitrate	25 Mbps
Video duration	3s - 60s
Video aspect ratio	0.01:1 a 10:6

CORRECAO AUTO (apenas type=POST): autoFitImage=true (padding) OU autoCropImage=true (crop). Mutuamente exclusivos.

7.2 TikTok - Videos

Constraint	Valor
Files	1 video (required)
Max size	1 GB
Resolution	360x360 a 4096x4096
Aspect ratio	0.01:1 a 10:6 (9:16 recomendado)
Duration	Max 10 min (600s)

Images (Photo Mode)

Constraint	Valor
Files	1-10 images
Format	JPG/JPEG/WebP ONLY (PNG REJEITADO!)
Max size	20 MB cada
Max width	1920px

Auto-scale: se foto > 1920x1080, redimensiona server-side. PNG = REJEITADO para Photo Mode.

7.3 YouTube - Shorts

Constraint	Valor
Files	1 video (required)
Max size	5 GB
Duration	Max 3 min (180s)
Aspect ratio	1:3 a 1:1 (vertical ou quadrado)

Long Form Videos

Constraint	Valor
Files	1 video (required)
Max size	5 GB
Duration	Max 4 horas (14.400s)
Aspect ratio	1:3 a 3:1

YouTube AUTO-DETECTA Shorts: duration < 3min E aspect vertical/quadrado. Nao ha botao "upload as Short".

7.4 Facebook - Page Posts

Constraint	Valor
Files	0-10 (imagens ou videos, nao misturados)
Max videos	1
Video max bitrate	45 Mbps
Video aspect ratio	0.01:1 a 1.91:1
Video duration	Max 20 min (1.200s)
Max images	4
Image max size	4 MB cada
Image aspect ratio	0.01:1 a 1.91:1

Reels

Constraint	Valor
Files	1 video (required)
Min resolution	540x960
Aspect ratio	0.01:1 a 10:6
Max bitrate	45 Mbps
Duration	3s - 20 min (1.200s)

Stories

Constraint	Valor
Files	1 (image or video)
Video min resolution	540x960
Video max bitrate	25 Mbps
Video duration	3s - 60s
Video aspect ratio	0.01:1 a 10:6
Image max size	4 MB
Image aspect ratio	0.01:1 a 10:6

7.5 LinkedIn

Constraint	Valor
Files	0-10 (imagens, videos, OU documentos - nao misturados)
Max videos	1
Video max size	2 GB
Video resolution	256x144 a 4096x4096
Video aspect ratio	1:2.4 a 2.4:1
Video duration	3s - 30 min
Max images	10
Image max size	5 MB cada
Image min resolution	200x200
Max documents	1
Document max size	100 MB
Document format	PDF only

7.6 Twitter / X

Constraint	Valor
Files	0-4 (imagens e/ou videos)
Video max size	512 MB
Video aspect ratio	1:3 a 3:1
Video min length	0.5s

Video duration (Free/Basic)	Max 140s (2:20)
Video duration (Premium+)	Max 600s (10 min)
Image max size	5 MB cada
Max images	4
Text (Free/Basic)	280 chars
Text (Premium+)	25.000 chars

7.7 Pinterest

Constraint	Valor
Files	1 (image or video, required)
Video max size	2 GB
Video aspect ratio	1:2 a 1.91:1
Video duration	4s - 15 min (900s)
Image max size	5 MB

7.8 Reddit - Single Post

Constraint	Valor
Files	0-1 (image or video)
Video max size	1 GB
Video aspect ratio	9:16 a 16:9
Video duration	4s - 15 min (900s)
Image max size	20 MB

Gallery Post

Constraint	Valor
Files	0-10 images (se subreddit permitir)
Image max size	20 MB cada

7.9 Threads

Constraint	Valor
Files	0-10 (imagens e/ou videos, misturados)
Video max size	1 GB
Video max bitrate	100 Mbps
Video aspect ratio	0.01:1 a 1.91:1
Image max size	8 MB
Image width	320-1440px
Image aspect ratio	0.01:1 a 1.91:1

7.10 Discord

Constraint	Valor
Files	0-10 (imagens e/ou videos, misturados)
Video max size	25 MB
Video aspect ratio	0.01:1 a 1.91:1
Image max size	25 MB
Image aspect ratio	0.01:1 a 1.91:1
Text	Max 2.000 chars

7.11 Slack

Constraint	Valor
Files	0-4 (imagens e/ou videos, misturados)
Video max size	1 GB
Video aspect ratio	0.01:1 a 1.91:1
Image max size	1 GB
Image aspect ratio	0.01:1 a 1.91:1
Text	Max 30.000 chars

7.12 Mastodon

Constraint	Valor
Files	0-4 (imagens e/ou videos, misturados)
Max videos	1
Video max size	99 MB
Video aspect ratio	1:3 a 1:1
Max images	4
Image max size	16 MB cada
Text	Depende da instancia (geralmente 500)

7.13 Bluesky

Constraint	Valor
Files	0-4 (imagens e/ou videos)
Text	300 chars

Bluesky tem restricoes minimas. bundle.social enforce apenas file count basico.

7.14 Google Business

Constraint	Valor
Files	0-1 image only (NO video)
Image max size	5 MB
Image min resolution	250x250
Text	Max 1.500 chars

7.15 Snapchat - Stories

Constraint	Valor
Files	Exatamente 1 image ou MP4 video
Max file size	100 MB
Video min resolution	540x960
Video duration	5-60 segundos

Spotlights

Constraint	Valor
Files	Exatamente 1 MP4 video
Max file size	100 MB
Min resolution	540x960
Video duration	6-60 segundos

7.16 Tabela Resumo

Plataforma	Max Files	Video Max	Video Dur	Img Max
IG Post	10	N/A(bitrate)	15 min	8 MB
IG Reel	1	N/A(bitrate)	15 min	-
IG Story	1	100 MB	60s	8 MB
TikTok Video	1	1 GB	10 min	-
TikTok Image	10	-	-	20 MB
YT Short	1	5 GB	3 min	-
YT Video	1	5 GB	4 horas	-
LinkedIn	10	2 GB	30 min	5 MB
Twitter/X	4	512 MB	140s/10min	5 MB
FB Post	10	N/A(bitrate)	20 min	4 MB
FB Reel	1	N/A(bitrate)	20 min	-
FB Story	1	N/A(bitrate)	60s	4 MB
Pinterest	1	2 GB	15 min	5 MB
Reddit	1/10	1 GB	15 min	20 MB
Threads	10	1 GB	-	8 MB
Discord	10	25 MB	-	25 MB
Slack	4	1 GB	-	1 GB
Mastodon	4	99 MB	-	16 MB
Bluesky	4	-	-	-
Google Biz	1	-	-	5 MB
Snap Story	1	100 MB	60s	100 MB
Snap Spot	1	100 MB	60s	-

7.17 Ceilings Globais

Video: 5 GB default (maior sob request, por org, via Multipart)

Imagens: 25 MB

Documentos (PDF): 100 MB

Formatos imagem: JPG, PNG, WEBP, GIF

Formatos video: MP4, MOV, WEBM

bundle.social enforce o MENOR entre ceiling global e limite da plataforma

8. FORMATO DE POST POR PLATAFORMA (15 redes)

8.1 TWITTER / X (data.TWITTER)

Campo	Tipo	Notas
text	string	~280 chars (maior se X Premium)
uploadIds	string[]	Ate 4 imagens OU 1 video/GIF
replySettings	enum	EVERYONE FOLLOWING MENTIONED_USERS SUBSCRIBERS VERIFIED
isAiGenerated	boolean	Label "made with AI" do X

REGRAS: Sem analytics. Sem set-channel (OAuth direto). Threads no X = primeiro tweet + comments API. firstComment NAO suportado.

8.2 INSTAGRAM (data.INSTAGRAM)

Campo	Tipo	Notas
type	enum	POST REEL STORY (default POST)
text	string	Caption. Max 2.000 chars
uploadIds	string[]	1 para POST/REEL/STORY. Carousel: 1-10 via carouselItems
altText	string	Alt text single image
carouselItems	array	[{uploadId, altText?, tagged?[]}]. Min 2, Max 10
thumbnailOffset	number(ms)	Frame do video como capa
thumbnail	string(URL)	Capa alternativa
shareToFeed	boolean	Reels: tambem no Feed
collaborators	string[]	Max 3 usernames, max 30 chars
tagged	array	[{username, x?(0-1), y?(0-1)}]. Image: x,y obrigatorios
locationId	string	ID via /misc/instagram/locations
autoFitImage	boolean	POST only. Padding para aspect ratio
autoCropImage	boolean	POST only. Crop para aspect ratio
trialParams	object	{graduationStrategy: MANUAL SS_PERFORMANCE}. Reels only
isPaidPartnership	boolean	Label "Paid partnership"
brandedContentSponsors	string[]	Max 2 usernames. FB Login only
musicSoundInfo	object	Reels only. {musicSoundId, musicSoundVolume?(0-100), videoOriginalSoundVolume?(0-100)}
isAiGenerated	boolean	Label AI Instagram

- REGRAS:
- OAuth via Meta Graph API. set-channel OBRIGATORIO
 - brandedContentSponsors e business discovery: apenas contas via Facebook Login
 - autoFitImage e autoCropImage: MUTUAMENTE EXCLUSIVOS. Apenas type=POST
 - musicSoundInfo: apenas type=REEL e conta via Facebook Login
 - carouselItems: cada uploadId deve existir em uploadIds, usar uma vez
 - tagged em carousel: usar carouselItems[].tagged (top-level rejeitado)
 - Video em carousel: person tags NAO suportados
 - firstComment maxLength: 2.200 chars

8.3 FACEBOOK (data.FACEBOOK)

Campo	Tipo	Notas
type	enum	POST REEL STORY (default POST)
text	string	Caption. Max 50.000 chars
uploadIds	string[]	Imagens ou 1 video
mediaItems	array	[[uploadId, altText?]]. Per-image alt text
link	string(URL)	Link attachment. type=POST only
mediaTitle	string	Titulo video. type=POST com video only. Max 255
thumbnail	string(URL)	Capa do video
nativeScheduleTime	ISO 8601	Agendar direto no Meta. Max 30 dias. Local time

REGRAS:

- OAuth via Meta Graph API (Page token). set-channel OBRIGATORIO (selecionar Page)
- Token expiration e a dor principal (Meta:190)
- Meta OAuth: selecionar TODAS as Pages que o usuario gerencia
- firstComment maxLength: 8.000 chars

8.4 TIKTOK (data.TIKTOK)

Campo	Tipo	Notas
type	enum	VIDEO IMAGE (default VIDEO)
text	string	Caption. Max 2.200 chars
uploadIds	string[]	1 video OU 1-10 imagens (Photo Mode)
thumbnail	string(URL)	Capa do video
privacy	enum	SELF_ONLY PUBLIC_TO_EVERYONE MUTUAL_FOLLOW_FRIENDS FOLLOWER_OF_CREATOR
photoCoverIndex	integer	Capa Photo Mode (zero-based, default 0)
thumbnailOffset	number(ms)	Frame do video como capa
isBrandContent	boolean	Parceria terceiro (paid)
isOrganicBrandContent	boolean	Promover proprio negocio
disableComments	boolean	Desabilitar comentarios
disableDuet	boolean	Desabilitar Duets
disableStitch	boolean	Desabilitar Stitches
isAiGenerated	boolean	Marcar como AI
autoAddMusic	boolean	TikTok adiciona musica em fotos
autoScale	boolean	IMAGE only. Redimensionar server-side
uploadToDraft	boolean	Salvar como draft (pode demorar horas!)
musicSoundInfo	object	{musicSoundId, musicSoundVolume?(0-100), musicSoundStart?(ms), musicSoundEnd?(ms), videoOriginalSoun

REGRAS:

- OAuth direto (sem set-channel)
- Photo Mode: JPG/JPEG/WebP ONLY. PNG = REJEITADO
- musicSoundId = song_clip_id do CML trending list
- privacy SELF_ONLY: permalink null
- REVIEW status: TikTok revisa (minutos a horas)
- uploadToDraft: pode causar delays massivos
- firstComment maxLength: 150 chars

8.5 YOUTUBE (data.YOUTUBE)

Campo	Tipo	Notas
type	enum	VIDEO SHORT (default SHORT)
uploadIds	string[]	1 video (required)
text	string	TITULO do video (required). Max 100 chars
description	string	Descricao. Max 5.000 chars
thumbnail	string(URL)	Capa custom (type=VIDEO only)
privacy	enum	PRIVATE PUBLIC UNLISTED (default PUBLIC)
defaultLanguage	string(BCP-47)	Idioma titulo/desc. Max 35 chars. Ex: en, pl
defaultAudioLanguage	string(BCP-47)	Idioma audio. Max 35 chars
madeForKids	boolean	REQUIRED por YouTube (COPPA/FTC)
containsSyntheticMedia	boolean	Conteudo AI
hasPaidProductPlacement	boolean	Paid placement

REGRAS:

- OAuth via YouTube Data API v3. set-channel OBRIGATORIO (canal)
- Shorts AUTO-DETECTADOS: duration < 3min + aspect vertical/quadrado
- madeForKids e REQUISITO LEGAL (FTC/COPPA)
- categoryId: via /misc/youtube/video-categories (nao vai no data)
- Playlists via /misc/youtube/playlist
- 10.000 units/dia (free tier Google)
- firstComment maxLength: 10.000 chars
- Monetizacao: requer withBusinessScope + rawYoutubeAnalyticsEnabled
- Dislikes disponiveis via API (escondidos no UI)

8.6 LINKEDIN (data.LINKEDIN)

Campo	Tipo	Notas
text	string(REQUIRED)	Max 3.000 chars. Perfil ou Company Page
uploadIds	string[]	Imagens, 1 video, ou 1 PDF
link	string(URL)	Article preview post
thumbnail	string(URL)	Capa
mediaTitle	string	Titulo video/documento. Max 200 chars
privacy	enum	CONNECTIONS PUBLIC LOGGED_IN CONTAINER
hideFromFeed	boolean	Nao mostrar no feed principal
disableReshare	boolean	Proibir resharing

REGRAS:

- OAuth via LinkedIn Marketing API. set-channel OBRIGATORIO (perfil ou Company Page)
- URNs para Company Pages
- Mentions: /misc/linkedin/mentions/builder
- firstComment maxLength: 1.250 chars
- Raw analytics: reaction-type breakdowns

8.7 THREADS (data.THREADS)

Campo	Tipo	Notas
text	string	~500 chars
uploadIds	string[]	Ate ~10 imagens ou 1 video
mediaItems	array	{{uploadId, altText?}}. Alternativa a uploadIds
topicTag	string	1-50 chars, sem pontos (.) ou ampersands (&)
replyControl	enum	everyone accounts_you_follow mentioned_only parent_post_author_only followers_only
linkAttachment	string(URL)	Link preview card. Text-only posts
poll	object	{optionA, optionB, optionC?, optionD?}. 2-4 opcoes, 1-25 chars. Text-only
gif	object	{gifId, provider?: GIPHY}. Text-only. Nao com poll
allowlistedCountryCodes	string[]	ISO 3166-1 alpha-2. Restringir visibilidade
crosspostToInstagramSto	boolean	Cross-post para IG Story
crosspostToInstagramSto	boolean	Cross-post em dark mode

- REGRAS:
- OAuth via Meta. Sem set-channel
 - Polls, GIFs, link attachments: APENAS text-only. Rejeita se uploadIds setado
 - poll e gif: mutuamente exclusivos
 - firstComment maxLength: 500 chars

8.8 BLUESKY (data.BLUESKY)

Campo	Tipo	Notas
text	string	~300 chars
uploadIds	string[]	Ate 4 imagens ou 1 video (set videoAlt)
tags	string[]	Hashtags sem #. Max 8
labels	enum[]	!no-unauthenticated porn sexual nudity graphic-media
quoteUri	string	AT-URI do post citado. ^at://AS+. Max 512
externalUrl	string(URI)	URL do link card
externalTitle	string	Titulo do card. Max 300
externalDescription	string	Descricao do card. Max 1.000
thumbnail	string(URI)	Imagem do card. Max 2048
videoAlt	string	Alt text do video. Max 10.000

- REGRAS:
- OAuth via AT Protocol. serverUrl opcional (default bsky.social)
 - Pode usar PDS/entryway custom
 - Sem set-channel
 - firstComment maxLength: 300 chars
 - Impressoes CALCULADAS (nao diretas)

8.9 MASTODON (data.MASTODON)

Campo	Tipo	Notas
text	string	Depende da instancia (geralmente ~500)
uploadIds	string[]	Ate 4 imagens ou 1 video
thumbnail	string(URL)	Capa do video
privacy	enum	PUBLIC UNLISTED PRIVATE DIRECT
spoiler	string	Content warning. Max 50 (instance pode ser menor)

REGRAS:

- OAuth via Mastodon API (instancia custom)
- serverUrl OBRIGATORIO no /connect
- Sem set-channel
- firstComment maxLength: 500 chars

8.10 PINTEREST (data.PINTEREST)

Campo	Tipo	Notas
boardName	string(REQUIRED)	Board para pinar
uploadIds	string[]	1 imagem (ou video)
text	string	Titulo do Pin. Max 100 chars
description	string	Descricao. Max 800 chars
link	string(URL)	URL para qual o Pin linka
altText	string	Alt text. Max 500 chars
note	string	Nota privada (nao visivel). Max 500 chars
thumbnail	string(URL)	Capa
dominantColor	string	Cor hex placeholder
isAiGenerated	boolean	Label AI Pinterest

REGRAS:

- OAuth via Pinterest API v5. Sem set-channel (boards refresh automatico)
- Sem comentarios via API
- Analytics suportado
- firstComment NAO suportado

8.11 REDDIT (data.REDDIT)

Campo	Tipo	Notas
sr	string(REQUIRED)	Subreddit: r/subredditName ou u/username
text	string(REQUIRED)	Titulo do post. Max 300 chars
description	string	Body para self post. Max 30.000 chars
uploadIds	string[]	1 imagem ou video. Gallery: ate 10
link	string(URL)	URL do link post
nsfw	boolean	Marcar NSFW
flairId	string	Obrigatorio se subreddit exigir. Via /misc/reddit/subreddit-flairs

REGRAS:

- OAuth via Reddit API. Sem set-channel
- ANTES de postar: /misc/reddit/post-requirements para ver limites
- firstComment maxLength: 10.000 chars
- Subreddit rules: nem todos permitem midia/galleries

8.12 DISCORD (data.DISCORD)

Campo	Tipo	Notas
channelId	string(REQUIRED)	ID do canal do server
text	string	Mensagem. Max 2.000 chars
uploadIds	string[]	Attachments
username	string	Nome exibido (webhook). Max 80 chars
avatarUrl	string(URL)	Avatar exibido. Max 2.048 chars

REGRAS: OAuth via webhook URL. Sem set-channel. Comentarios = mensagens. Sem analytics. firstComment maxLength: 2.000 chars.

8.13 SLACK (data.SLACK)

Campo	Tipo	Notas
channelId	string(REQUIRED)	ID do canal do workspace
text	string	Mensagem. Max 30.000 chars
uploadIds	string[]	Attachments
username	string	Nome exibido. Max 80 chars
avatarUrl	string(URL)	Avatar exibido. Max 2.048 chars

REGRAS: OAuth via webhook URL. Sem set-channel. Comentarios = mensagens. Sem analytics. firstComment maxLength: 30.000 chars.

8.14 GOOGLE_BUSINESS (data.GOOGLE_BUSINESS)

Campo	Tipo	Notas
text	string	Corpo do update. Max 1.500 chars
uploadIds	string[]	Imagens (1 imagem only, sem video)
topicType	enum	STANDARD EVENT OFFER ALERT (default STANDARD)
languageCode	string	Tag idioma: en, en-US
callToActionType	enum	BOOK ORDER SHOP LEARN_MORE SIGN_UP CALL
callToActionUrl	string(URL)	URL do CTA
eventTitle	string	topicType=EVENT. Max 58 chars
eventStartDate	ISO 8601	topicType=EVENT
eventEndDate	ISO 8601	topicType=EVENT
offerCouponCode	string	topicType=OFFER. Max 58 chars
offerRedeemOnlineUrl	string	topicType=OFFER
offerTermsConditions	string	topicType=OFFER. Max 1.500 chars
alertType	enum	COVID_19 (default). topicType=ALERT

REGRAS:

- OAuth via Google My Business API. set-channel OBRIGATORIO (local)
- Sem comentarios via API
- Analytics suportado
- Reviews via /misc/google-business/reviews
- Location management via /misc/google-business/location
- firstComment NAO suportado

8.15 SNAPCHAT (data.SNAPCHAT)

Campo	Tipo	Notas
type	enum	STORY SPOTLIGHT (default STORY)
uploadIds	string[]	1 imagem ou video. Spotlight: 1 MP4 only
text	string	Alias para description (Spotlight). Max 160 chars
description	string	Spotlight description. Max 160 chars
locale	string	Spotlight locale. en_US. Default en_US
skipSaveToProfile	boolean	Spotlight only. Nao salvar no perfil

REGRAS:

- OAuth via Snapchat Marketing API (snapkit.com). Sem set-channel
- Requer Snapchat Public Profile
- Sem comentarios via API
- Analytics suportado (profile + content)
- firstComment NAO suportado

9. LIMITES DE TEXTO E CARACTERES

Plataforma	Main text	Outros campos
Instagram	text: 2.000	collaborators max 30, tagged max 30, altText opcional
TikTok	text: 2.200	-
YouTube	text (titulo): 100	description: 5.000
LinkedIn	text: 3.000	mediaTitle: 200
Twitter/X	text: 280 (Free) / 25.000 (Premium+)	-
Facebook	text: 50.000	mediaTitle: 255, link: 2.048, altText opcional
Pinterest	text: 100	description: 800, altText: 500, note: 500
Reddit	text (titulo): 300	description: 30.000
Threads	text: 500	altText opcional em image media
Discord	text: 2.000	username: 80, avatarUrl: 2.048
Slack	text: 30.000	username: 80, avatarUrl: 2.048
Mastodon	text: 30.000	spoiler: 50 (instance pode ser menor)
Bluesky	text: 300	externalTitle: 300, externalDescription: 1.000, videoAlt: 10.000
Google Business	text: 1.500	eventTitle: 58, offerCouponCode: 58, offerTerms: 1.500
Snapchat	description/text: 160	locale: en_US

First Comment Limits (maxLength)

Plataforma	maxLength
TIKTOK	150
YOUTUBE	10.000
INSTAGRAM	2.200
FACEBOOK	8.000
THREADS	500
LINKEDIN	1.250
REDDIT	10.000
MASTODON	500
DISCORD	2.000
SLACK	30.000
BLUESKY	300

TWITTER, PINTEREST, GOOGLE_BUSINESS, SNAPCHAT: firstComment NAO suportado.

10. RATE LIMITS

10.1 API Rate Limits (3 camadas simultaneas)

Camada	Janela	Max	Equiv	Previne
Burst	1 segundo	100	6.000/min	Promise.all() com 500 req
Short	10 segundos	500	3.000/min	Polling agressivo, loops
Minute	1 minuto	2.000	2.000/min	Abuso volume alto

Tracking: por ENDPOINT + por tracker (API key, bearer token, ou IP). Trafego em POST /post NAO consome bucket de GET /analytics. Exceder = 429.

10.2 Daily Post Limits por Plataforma (por conta real)

Plataforma	FREE	PRO	BUSINESS
TWITTER	5	15	15
FACEBOOK	10	50	100
INSTAGRAM	10	50	100
LINKEDIN	10	18	24
YOUTUBE	10	10	15
TIKTOK	5	10	15
THREADS	10	200	250
PINTEREST	10	24	36
REDDIT	10	24	36
DISCORD	10	100	200
SLACK	10	100	200
MASTODON	10	50	100
BLUESKY	10	50	100
GOOGLE_BUSINESS	10	20	40
SNAPCHAT	5	20	40

REGRA CHAVE: Cota segue a CONTA REAL (platform + account_id), NAO a conexao. Mesma conta em 5 teams = cota total compartilhada. Contagem: por UTC calendar day.

10.3 Daily Comment Limits

Plataforma	FREE	PRO	BUSINESS
FACEBOOK	3	50	100
INSTAGRAM	3	50	100
LINKEDIN	3	18	24
YOUTUBE	3	10	15
TIKTOK	3	10	15
THREADS	3	200	250
REDDIT	3	24	36
DISCORD	3	100	200
SLACK	3	100	200
MASTODON	3	50	100
BLUESKY	3	50	100

Comments API: FB, IG, LI, YT, TT, Reddit, Threads, Mastodon, Discord, Slack, Bluesky. Twitter/X, Pinterest, Google Business NAO suportados.

10.4 Monthly Organization Caps

Tier	Posts/mes	Comments/mes	Uploads/mes
FREE	20	50	200

PRO	10.000	5.000	100.000
BUSINESS	100.000	50.000	1.000.000

Contagem: por data de CRIACAO no mes UTC. Rascunhos nao contam. Falhas NAO devolvem cota. Reset: dia 1 UTC.

10.5 Import Limits

Tipo	Limite	Reset
Post history	FREE:5/PRO:100/BUSINESS:500 por conta/mes	Dia 1 UTC
Post history request	100 posts por request (hard cap)	-
Comment import	FREE:25/PRO:200/BUSINESS:1.000 por post/import	Sem janela
Review import	FREE:5/PRO:200/BUSINESS:200 por conta/mes	Dia 1 UTC
Review request	250 reviews por request (hard cap)	-

10.6 Force Refresh Analytics

Formula: max force refreshes/dia = numero_de_teams x 5 (por org). Ex: 10 teams = 50/dia. Exceder = 429.

11. WEBHOOKS

11.1 Eventos

Evento	Quando dispara
post.published	Post terminou (POSTED ou ERROR). Checar data.status
comment.published	Comentario terminou (POSTED ou ERROR)
social-account.created	Usuario conectou nova conta
social-account.updated	Conta mudou (canal removido, desconexao remota)
social-account.deleted	Conta desconectada/removida
team.created	Novo team criado
team.updated	Detalhes do team mudaram (inclui add/remove contas)
team.deleted	Team deletado

11.2 Headers

Header	Valor
Content-Type	application/json
User-Agent	bundlesocial
x-signature	HMAC-SHA256 (verificar com Signing Secret)

11.3 Payload post.published

```
{
  "type": "post.published",
  "data": {
    "id": "post_abc123",
    "title": "My post",
    "status": "POSTED",
    "postDate": "2026-01-15T10:00:00.000Z",
    "postedDate": "2026-01-15T10:00:02.341Z",
    "teamId": "team_xyz",
    "organizationId": "org_123",
    "data": { "INSTAGRAM": { "type": "POST", "text": "Hello!", "uploadIds": ["upload_456"] } },
    "error": null,
    "errorsVerbose": null,
    "externalData": {
      "INSTAGRAM": { "id": "17900000000000000", "permalink": "https://instagram.com/p/example/" }
    },
    "retryCount": 0,
    "uploads": [{ "upload": { "id": "upload_456", "fileName": "cover.jpg", "mimeType": "image/jpeg", "size": 245000 } }],
    "socialAccounts": [{ "socialAccount": { "id": "sa_789", "type": "INSTAGRAM", "username": "mybrand" } }]
  }
}
```

accessToken, refreshToken, secret, expiresAt NUNCA aparecem em webhooks.

11.4 Delivery e Retries

Config	Valor
Timeout	15 segundos por request
Max tentativas	3 (inicial + 2 retries)
Backoff	Exponencial, começando em 30 segundos
Concorrencia	Ate 50 deliveries simultaneos
Auto-disable	7 dias sem sucesso = desativado
Re-enable	Dashboard ou API. Reset contador na proxima entrega OK
Eventos perdidos	NAO sao replayed apos disable

Resend manual Dashboard ou API. Fresh delivery com mesmo payload

11.5 Remote Disconnection Detection

Checa cada conta a cada 6 HORAS. Cobertura: Meta (FB, IG, Threads). Outras: /social-account/connection-check manual.

FLUXO:

- 1. Validar token a cada 6h
- 2. Se falhar: 3 retries com 10min backoff
- 3. Se 3 falhas: agendar delecao + webhook social-account.updated
- 4. Grace period 6h antes de deletar
- 5. Delecao definitiva: webhook social-account.deleted

12. SISTEMA DE ERROS PADRONIZADO

12.1 Estrutura

```
{
  "errorsVerbose": {
    "INSTAGRAM": {
      "code": "META:190",
      "userFacingMessage": "Your Instagram token expired. Please reconnect.",
      "errorMessage": "OAuthException: Validate permission failed.",
      "isTransient": false
    },
    "TIKTOK": null
  }
}
```

12.2 Campos

Campo	Significado
code	ID estavel (ex: HTTP:429). Usar para logica
userFacingMessage	Seguro mostrar ao usuario
errorMessage	Logs raw developer
isTransient	true = retry. false = corrigir input/auth

12.3 Prefixos

Prefixo	Plataforma
META	Instagram, Facebook, Threads
TT	TikTok
LI	LinkedIn
YT	YouTube
HTTP	Erros genericos

12.4 Erros Canonicos Comuns (Meta)

Error key	Platform code	Causa
account_connection_expired	HTTP:401, Meta:190:460, Meta:190:467	Token/sessao invalidado
page_or_target_inaccessible	HTTP:400, Meta:190	Page nao acessivel
media_fetch_failed	HTTP:422	Crawler bloqueado (403)
account_authorization_revoked	Meta:190:458	App authorization removida
posting_rate_limited	Meta:368:1390008	Limite de frequencia/spam
identity_confirmation_required	Meta:368:4854002	Confirmar identidade no app
account_not_confirmed	HTTP:401, Meta:190:464	Conta nao confirmada
missing_platform_permission	HTTP:400, Meta:190, Meta:200	Permissoes de Page revogadas
account_security_limited	HTTP:400, Meta:368:1404112	Conta limitada por seguranca
platform_login_required	HTTP:401, Meta:190:459	Checkpoint de login
page_role_or_2fa_required	Meta:190:492	Page role ou 2FA necessario
request_too_large	Meta:1	Request com dados demais

13. ANALYTICS

13.1 Refresh

Default: automatico a cada 24 HORAS (sem cron job do cliente).
Force refresh: POST /analytics/post/force ou /analytics/social-account/force
Force rate limit: teams x 5 por dia (por org). Exceder = 429.
Custom refresh period: negociavel (ex: 15min, custo extra).

13.2 Data Retention

Analytics retidos por 30 DIAS. Apos isso, deletados. Para historico longo: cron job diario que busca e salva no seu banco.

13.3 Tipos de Janela

Tipo	Descricao
Rolling Window	Dados agregados em periodo fixo (ex: ultimos 30 dias). Atualiza conforme tempo passa.
Lifetime	Totais cumulativos desde criacao. Valores geralmente so aumentam.

13.4 Metricas Normalizadas

impressions | impressionsUnique | views | viewsUnique | likes | comments | shares | saves | postCount | followers | following

13.5 Especificidades por Plataforma

TikTok - Profile (Rolling 30 dias)

impressions=views, likes (agregado), comments, postCount, followers, following. Demograficos pode levar 48h.

TikTok - Post (Lifetime)

impressions=views, likes, comments, shares. saves=0 (nao fornecido).

YouTube - Profile (Lifetime)

impressions=views (nao distingue unique), likes=0 (canal), comments, postCount=subscribers, following=0.

YouTube - Post (Lifetime)

impressions=views, likes, dislikes (via API!), comments, shares=0, saves=favorites. Demograficos requer rawYoutubeAnalyticsEnabled.

YouTube - Monetizacao

Requer: rawYoutubeAnalyticsEnabled + withBusinessScope no OAuth. Metricas: estimatedRevenue, estimatedAdRevenue, grossRevenue, cpm, adImpressions, monetizedPlaybacks.

13.6 Raw Analytics por Plataforma

Plataforma	Profile raw	Post raw	Conteudo tipico
Instagram	Sim	Sim	Account info, daily totals, demographics, media insights
Facebook	Sim	Sim	Page info, insights, demographic, reaction/video metrics
LinkedIn	Sim	Sim	Member/org analytics, share stats, reaction breakdowns
TikTok	Sim	Sim	Account metrics, audience splits, video watch/source
YouTube	Sim	Sim	Channel/video metrics, demographics, monetization
Snapchat	Sim	Sim	Profile + content analytics
Threads	Sim	Sim	Profile + post raw
Pinterest	Sim	Sim	Profile + Pin raw
Reddit	Sim	Sim	Profile + post raw
Mastodon	Sim	Sim	Basic metrics
Bluesky	Sim	Sim	Calculated impressions, limited data
Google Business	Sim	Sim	Search queries, direction requests, calls

14. DATA RETENTION

Tipo de dado	Retencao
Analytics (parsed + raw)	30 dias
Webhook events	7 dias
Uploads deletados	7 dias apos delecao
Uploads nao usados (sem posts)	Soft-delete 90 dias + 7 dias purge
Multipart nao completado	Auto-abort 7 dias
Posts importados (history)	30 dias
Comentarios importados	Permanente (truncado ao limite na importacao)

REGRA DE OURO: Se precisa de historico longo de analytics, armazene voce mesmo. Cron job diario que busca e salva no seu banco.

15. OAUTH E CHANNEL SELECTION

Plataforma	OAuth enough?	set-channel?	Notas
Facebook	Nao	Sim	Selecionar Facebook Page
Instagram via FB	Nao	Sim	Selecionar conta IG conectada a Page
Instagram direto	Sim	Nao	instagramConnectionMethod: INSTAGRAM
LinkedIn	Nao	Sim	Perfil ou Company Page
YouTube	Nao	Sim	Selecionar canal
Snapchat	Sim	Nao	Usa Public Profile do OAuth
Google Business	Nao	Sim	Selecionar business location
Pinterest	Geralmente	Nao	Boards via refresh-channels
Reddit	Geralmente	Nao	Destinations via refresh-channels
Slack	Geralmente	Nao	Webhook channels via refresh-channels
Discord	Geralmente	Nao	Webhook channels via refresh-channels
TikTok	Sim	Nao	OAuth direto
Twitter/X	Sim	Nao	OAuth direto
Threads	Sim	Nao	Meta infra, OAuth direto
Mastodon	Sim	Nao	serverUrl obrigatorio no /connect
Bluesky	Sim	Nao	serverUrl opcional (default bsky.social)

OAuth Options Especiais

disableAutoLogin: evita login automatico

tiktokForceLogin: TikTok only. Rota via logout para fresh login

forceBrowserOAuth: forca fluxo browser

withBusinessScope: YouTube only. Concede monetary analytics scope

instagramConnectionMethod: INSTAGRAM (direto) ou FACEBOOK (via Meta)

Meta Rule (CRITICO)

Para Facebook e Instagram via Facebook:

1. Pedir usuario para selecionar TODAS as Pages e contas IG no OAuth Meta
2. Apos OAuth, selecionar UMA Page/conta para o team (set-channel)
3. Meta mantem grant para todas as Pages, bundle.social armazena target separado

PROBLEMA: Se usuario conectar multiplas teams e selecionar so uma Page em cada OAuth, Meta pode sobrescrever grants. Uma team parece conectada mas Meta removeu grant. Publishing falha com: "must be granted before impersonating a user's page".

SOLUCAO: Reconnect -> selecionar TODAS as Pages -> escolher 1 por team.

16. COMMENTS API

16.1 Criar Comentario

POST /api/v1/comment/

Body:

```
{
  "teamId": "team_123",
  "title": "string (min 1)",
  "internalPostId": "post_abc",
  "internalParentCommentId": null,
  "fetchParentCommentId": null,
  "postDate": "2026-09-01T15:00:00.000Z",
  "status": "DRAFT | SCHEDULED",
  "socialAccountTypes": ["INSTAGRAM", "TIKTOK"],
  "data": {
    "FACEBOOK": { "text": "..." },
    "INSTAGRAM": { "text": "..." },
    "TIKTOK": { "text": "..." },
    "YOUTUBE": { "text": "..." },
    "THREADS": { "text": "..." },
    "LINKEDIN": { "text": "..." },
    "REDDIT": { "text": "..." },
    "MASTODON": { "text": "..." },
    "DISCORD": { "text": "..." },
    "SLACK": { "text": "..." },
    "BLUESKY": { "text": "..." }
  }
}
```

Suportados: TIKTOK, YOUTUBE, INSTAGRAM, FACEBOOK, THREADS, LINKEDIN, REDDIT, MASTODON, DISCORD, SLACK, BLUESKY

NAO suportados: TWITTER, PINTEREST, GOOGLE_BUSINESS, SNAPCHAT

16.2 Importar Comentarios (async)

POST /api/v1/comment/import

Body: { teamId, postId ou importedPostId, socialAccountType }

Plataformas: FB, IG, LI, YT, TT, Reddit, Threads, Mastodon, Bluesky

Job assincrono. Status: queued -> running -> finished/failed.

Limite: FREE 25 / PRO 200 / BUSINESS 1.000 comentarios por post/import.

GET /comment/import -> listar

GET /comment/import/{importId} -> status

GET /comment/import/comments -> comentarios importados (paginado)

POST /comment/import/comments/{commentId}/action -> acao

16.3 First Comment (automatico)

Definido no POST /post em firstComment.PLATFORM.

Agendado apos publicacao bem-sucedida.

Cada plataforma tem maxLength (ver secao 9).

TWITTER, PINTEREST, GOOGLE_BUSINESS, SNAPCHAT: nao suportado.

17. IMPORTS

17.1 Post History Import (async)

POST /api/v1/post-history-import/

Body: { teamId, socialAccountType, ... }

Job assincrono puxa posts existentes da plataforma.

Backfill do dashboard com historico anterior a conexao.

Limites mensais por conta:

FREE: 5 posts/conta/mes

PRO: 100 posts/conta/mes

BUSINESS: 500 posts/conta/mes

Hard cap por request: 100 posts

GET /post-history-import/ -> listar

GET /post-history-import/{importId} -> status

POST /post-history-import/{importId}/retry -> re-tentar

GET /post-history-import/posts -> posts importados

DELETE /post-history-import/posts -> deletar

Posts importados retidos por 30 dias.

17.2 Bulk Post from CSV (async)

POST /api/v1/post-csv-import/

Upload CSV com multiplos posts.

Processamento assincrono com resultados por linha.

GET /post-csv-import/ -> listar

GET /post-csv-import/{importId} -> status

GET /post-csv-import/{importId}/rows -> linhas

GET /post-csv-import/{importId}/status -> status detalhado

17.3 Google Reviews / Facebook Recommendations Import

POST /misc/google-business/reviews/import

POST /misc/facebook/recommendations/import

Limites mensais por conta:

FREE: 5 | PRO: 200 | BUSINESS: 200 por conta/mes

Hard cap por request: 250

Responder reviews:

PUT /misc/google-business/reviews/{reviewId}/reply

DELETE /misc/google-business/reviews/{reviewId}/reply

PUT /misc/facebook/recommendations/{id}/reply

18. MISC ENDPOINTS (operacoes por plataforma)

18.1 Facebook

Endpoint	Funcao
PATCH /misc/facebook/comment	Editar comentario
DELETE /misc/facebook/comment	Deletar comentario
PATCH /misc/facebook/post	Editar post
DELETE /misc/facebook/post	Deletar post
GET /misc/facebook/recommendations	Listar recommendations
POST /misc/facebook/recommendations/import	Importar recommendations
GET /misc/facebook/recommendations/{id}	Detalhes
PUT /misc/facebook/recommendations/{id}/reply	Responder
GET /misc/facebook/token-debug	Debug token Meta

18.2 Instagram

Endpoint	Funcao
GET /misc/instagram/audio	Buscar audio para Reels
DELETE /misc/instagram/comment	Deletar comentario
GET /misc/instagram/locations	Buscar locations
GET /misc/instagram/tags	Business discovery

18.3 TikTok

Endpoint	Funcao
GET /misc/tiktok/cml/trending-list	Trending music (song_clip_id)
DELETE /misc/tiktok/comment	Deletar comentario

18.4 YouTube

Endpoint	Funcao
PATCH/DELETE /misc/youtube/comment	Editar/deletar comentario
GET/POST/PUT/DELETE /misc/youtube/playlist	CRUD playlists
POST/GET/DELETE /misc/youtube/playlist-items	Items de playlist
POST /misc/youtube/thumbnail	Set custom thumbnail
PATCH/DELETE /misc/youtube/video	Atualizar/deletar video
GET /misc/youtube/regions	Listar regioes
GET /misc/youtube/video-categories	Listar categorias (categoryId)

18.5 LinkedIn

Endpoint	Funcao
PATCH/DELETE /misc/linkedin/comment	Editar/deletar comentario
POST /misc/linkedin/mentions/builder	Construir mentions (URNs)
GET /misc/linkedin/mentions/tags	Tags de mentions
PATCH/DELETE /misc/linkedin/post	Editar/deletar post
POST /misc/linkedin/post/reshare	Reshare post

18.6 Reddit

Endpoint	Funcao
PATCH/DELETE /misc/reddit/comment	Editar/deletar comentario

PATCH/DELETE /misc/reddit/post	Editar/deletar post
GET /misc/reddit/post-requirements	Regras do subreddit (ANTES de postar)
GET /misc/reddit/subreddit-flairs	Flairs disponiveis (flairId)

18.7 Google Business

Endpoint	Funcao
GET/PATCH /misc/google-business/location	Dados/atualizar localizacao
GET/PATCH /misc/gb/location/attributes	Atributos
GET /misc/gb/location/attributes/available	Atributos disponiveis
GET /misc/gb/location/categories/available	Categorias disponiveis
GET/PATCH /misc/gb/location/food-menus	Cardapios
PATCH /misc/gb/location/hours	Horarios
GET/POST/PATCH/DELETE /misc/gb/location/place-action-links	Links de acao
GET/PATCH /misc/gb/location/service-list	Lista de servicos
POST/GET/DELETE /misc/google-business/media	Midia
DELETE /misc/google-business/post	Deletar post
GET /misc/google-business/reviews	Listar reviews
POST /misc/google-business/reviews/import	Importar reviews (async)
GET /misc/gb/reviews/{reviewId}	Detalhes review
PUT/DELETE /misc/gb/reviews/{reviewId}/reply	Responder/remover resposta

18.8 Outras

Plataforma	Operacoes
Twitter/X	DELETE /misc/twitter/tweet
Pinterest	PATCH/DELETE /misc/pinterest/pin
Mastodon	PATCH/DELETE /misc/mastodon/comment, PATCH/DELETE /misc/mastodon/status
Bluesky	DELETE /misc/bluesky/comment, DELETE /misc/bluesky/post
Discord	DELETE /misc/discord/message
Slack	PATCH/DELETE /misc/slack/message

19. BLUEPRINT POSTGRESQL (schema completo)

```
-- Hierarquia: Organization > Team > SocialAccount

CREATE TABLE organizations (
  id TEXT PRIMARY KEY DEFAULT 'org_' || gen_random_uuid(),
  name TEXT NOT NULL,
  plan TEXT DEFAULT 'FREE',
  subscription_status TEXT,
  raw_youtube_analytics_enabled BOOLEAN DEFAULT FALSE,
  created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE teams (
  id TEXT PRIMARY KEY DEFAULT 'team_' || gen_random_uuid(),
  organization_id TEXT REFERENCES organizations(id) ON DELETE CASCADE,
  name TEXT NOT NULL,
  avatar_url TEXT,
  created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE api_keys (
  id TEXT PRIMARY KEY DEFAULT 'key_' || gen_random_uuid(),
  organization_id TEXT REFERENCES organizations(id) ON DELETE CASCADE,
  key_hash TEXT NOT NULL,
  prefix TEXT,
  last_used_at TIMESTAMPTZ,
  created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE social_accounts (
  id TEXT PRIMARY KEY DEFAULT 'sa_' || gen_random_uuid(),
  team_id TEXT REFERENCES teams(id) ON DELETE CASCADE,
  type TEXT NOT NULL,
  username TEXT, display_name TEXT, external_id TEXT,
  avatar_url TEXT,
  access_token TEXT,
  refresh_token TEXT,
  token_expires_at TIMESTAMPTZ,
  channel_id TEXT, channel_data JSONB,
  status TEXT DEFAULT 'ACTIVE',
  scheduled_for_deletion_at TIMESTAMPTZ,
  created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE posts (
  id TEXT PRIMARY KEY DEFAULT 'post_' || gen_random_uuid(),
  team_id TEXT REFERENCES teams(id) ON DELETE CASCADE,
  title TEXT, status TEXT DEFAULT 'DRAFT',
  post_date TIMESTAMPTZ, posted_date TIMESTAMPTZ,
  data JSONB NOT NULL,
  first_comment JSONB,
  external_data JSONB,
  error TEXT, errors_verbose JSONB,
  retry_count INT DEFAULT 0,
  reference_key TEXT,
  created_at TIMESTAMPTZ DEFAULT NOW(),
  updated_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE uploads (
```

```

    id TEXT PRIMARY KEY DEFAULT 'upload_' || gen_random_uuid(),
    team_id TEXT REFERENCES teams(id) ON DELETE CASCADE,
    file_name TEXT NOT NULL, mime_type TEXT NOT NULL,
    size BIGINT, storage_path TEXT NOT NULL,
    status TEXT DEFAULT 'ACTIVE', deleted_at TIMESTAMPTZ,
    created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE comments (
    id TEXT PRIMARY KEY DEFAULT 'comment_' || gen_random_uuid(),
    team_id TEXT REFERENCES teams(id) ON DELETE CASCADE,
    internal_post_id TEXT REFERENCES posts(id),
    internal_parent_comment_id TEXT,
    fetched_parent_comment_id TEXT,
    title TEXT, status TEXT, post_date TIMESTAMPTZ,
    data JSONB, external_data JSONB,
    error TEXT, errors_verbose JSONB,
    retry_count INT DEFAULT 0,
    created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE webhooks (
    id TEXT PRIMARY KEY DEFAULT 'wh_' || gen_random_uuid(),
    organization_id TEXT REFERENCES organizations(id) ON DELETE CASCADE,
    url TEXT NOT NULL, signing_secret TEXT NOT NULL,
    status TEXT DEFAULT 'ACTIVE',
    last_success_at TIMESTAMPTZ,
    failure_count INT DEFAULT 0,
    created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE webhook_events (
    id TEXT PRIMARY KEY DEFAULT 'ev_' || gen_random_uuid(),
    webhook_id TEXT REFERENCES webhooks(id) ON DELETE CASCADE,
    type TEXT NOT NULL, payload JSONB NOT NULL,
    status TEXT, response_code INT, attempts INT DEFAULT 0,
    next_retry_at TIMESTAMPTZ,
    created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE analytics_snapshots (
    id TEXT PRIMARY KEY DEFAULT 'an_' || gen_random_uuid(),
    social_account_id TEXT REFERENCES social_accounts(id) ON DELETE CASCADE,
    post_id TEXT REFERENCES posts(id) ON DELETE CASCADE,
    platform_type TEXT NOT NULL,
    metrics JSONB NOT NULL,
    raw JSONB,
    forced BOOLEAN DEFAULT FALSE,
    fetched_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE usage_counters (
    id TEXT PRIMARY KEY DEFAULT 'uc_' || gen_random_uuid(),
    organization_id TEXT REFERENCES organizations(id),
    team_id TEXT REFERENCES teams(id),
    counter_type TEXT NOT NULL,
    period TEXT NOT NULL,
    used INT DEFAULT 0,
    UNIQUE(team_id, counter_type, period)
);

```

20. STACK RECOMENDADA E ARQUITETURA

Camada	bundle.social	Recomendado (superior)
Backend	Node.js + Express/Nest	Node.js + NestJS + Fastify
Linguagem	TypeScript	TypeScript (strict)
Banco	PostgreSQL	PostgreSQL 16 + Prisma/Drizzle
Cache	Redis	Redis 7 + Upstash (serverless)
Fila	BullMQ/SQS	BullMQ + Redis ou Temporal
Storage	S3-compativel	Cloudflare R2 (zero egress)
Frontend	Next.js	Next.js 15 + App Router + Server Components
Auth	API Key	API Key + JWT opcional + RBAC granular
Docs	Mintlify/custom	Mintlify ou Fuma (com llms.txt)
Spec	OpenAPI 3.0	OpenAPI 3.1 + Zod schemas
SDK	TypeScript	TS + Python + Go (gerar de OpenAPI)
Monitoring	BetterStack	Grafana + Loki + Prometheus
Deploy	Desconhecido	Docker + Coolify ou Kubernetes
CDN	Cloudflare	Cloudflare (free tier)
MCP	Sim	Sim + A2A (Agent-to-Agent)

Interface PlatformAdapter (padrao Strategy)

```
interface PlatformAdapter {
  getAuthUrl(state: string, options?: OAuthOptions): string;
  handleCallback(code: string): Promise<TokenSet>;
  refreshToken(refreshToken: string): Promise<TokenSet>;
  getChannels(tokenSet: TokenSet): Promise<Channel[]>;
  publish(post: PostData, account: SocialAccount): Promise<PublishResult>;
  getAnalytics(account: SocialAccount, postId?: string): Promise<Analytics>;
  validateConnection(account: SocialAccount): Promise<boolean>;
}

// 15 implementacoes necessarias:
// InstagramAdapter (Meta Graph API)
// FacebookAdapter (Meta Graph API, Page token)
// TikTokAdapter (TikTok Content API)
// YouTubeAdapter (YouTube Data API v3)
// LinkedInAdapter (LinkedIn Marketing API)
// TwitterAdapter (X API v2)
// ThreadsAdapter (Threads API)
// BlueskyAdapter (AT Protocol)
// MastodonAdapter (Mastodon API, instancia custom)
// PinterestAdapter (Pinterest API v5)
// RedditAdapter (Reddit API)
// DiscordAdapter (webhook URL)
// SlackAdapter (webhook URL)
// GoogleBusinessAdapter (Google My Business API)
// SnapchatAdapter (Snapchat Marketing API)
```

Rate Limiter (3 camadas com Redis)

```
async function rateLimiter(req, reply, done) {
  const tracker = req.headers['x-api-key'] || req.ip;
  const endpoint = `${req.method}:${req.routeOptions.url}`;
  const windows = [
    { key: `${tracker}:${endpoint}:1s`, limit: 100, ttl: 1 },
    { key: `${tracker}:${endpoint}:10s`, limit: 500, ttl: 10 },
    { key: `${tracker}:${endpoint}:60s`, limit: 2000, ttl: 60 },
  ];
};
```

```
for (const w of windows) {
  const count = await redis.incr(w.key);
  if (count === 1) await redis.expire(w.key, w.ttl);
  if (count > w.limit) {
    return reply.code(429).send({
      error: 'RATE_LIMITED', window: w.ttl + 's', limit: w.limit
    });
  }
}
done();
}
```

21. CRON JOBS E BACKGROUND WORKERS

Job	Frequencia	Funcao
Analytics refresh	A cada 24h	Buscar analytics de todas as contas
Connection check (Meta)	A cada 6h	Validar tokens Meta (FB, IG, Threads)
Token refresh	Proativo	Renovar tokens prestes a expirar
Stale upload cleanup	Diario	Soft-delete uploads nao usados ha 90 dias
Deleted upload purge	Diario	Remover fisicamente uploads deletados ha 7 dias
Analytics purge	Diario	Deletar analytics com 30 dias
Webhook events purge	Diario	Deletar eventos com 7 dias
Imported posts purge	Diario	Deletar posts importados com 30 dias
Monthly counter reset	Dia 1 (UTC)	Zerar contadores mensais de uso
Webhook auto-disable	Diario	Desativar webhooks sem sucesso ha 7 dias
Multipart auto-abort	Diario	Abortar multipart nao completado ha 7 dias

Webhook Delivery Worker

```
async function webhookWorker(event) {
  const webhook = await getWebhook(event.webhookId);
  if (webhook.status === 'DISABLED') return;
  const payload = JSON.stringify({ type: event.type, data: event.payload });
  const signature = hmacSHA256(payload, webhook.signing_secret);
  try {
    const res = await fetch(webhook.url, {
      method: 'POST',
      headers: { 'Content-Type': 'application/json', 'User-Agent': 'yourapp', 'x-signature': signature },
      body: payload,
      signal: AbortSignal.timeout(15000)
    });
    if (res.ok) { await markDelivered(event.id); await resetFailureCount(webhook.id); }
    else { throw new Error(`HTTP ${res.status}`); }
  } catch (err) {
    await incrementAttempts(event.id);
    if (event.attempts < 3) {
      const delay = 30 * Math.pow(3, event.attempts - 1);
      await scheduleRetry(event.id, delay);
    } else {
      await markFailed(event.id);
      await incrementFailureCount(webhook.id);
      if (daysSinceLastSuccess(webhook.id) >= 7) {
        await disableWebhook(webhook.id);
        await notifyOwner(webhook.organizationId);
      }
    }
  }
}
```

22. ROADMAP DE IMPLEMENTACAO

Fase 1: MVP (4-6 semanas)

1. Setup: NestJS + Prisma + PostgreSQL + Redis + R2
2. Auth: API Key + Organization/Team hierarchy
3. Upload: Simple + Direct (sem multipart ainda)
4. 3 plataformas: Instagram, Twitter/X, LinkedIn
5. POST /post (sem firstComment, sem scheduling nativo)
6. Webhooks basicos (post.published)
7. Analytics basico (parsed, sem raw)
8. Dashboard minimo (Next.js)

Fase 2: Core (8-12 semanas)

1. Multipart upload (videos grandes)
2. +6 plataformas: TikTok, YouTube, Facebook, Threads, Pinterest, Reddit
3. Scheduling completo (BullMQ)
4. firstComment automatico
5. Comments API (criar + importar)
6. Rate limits completos (3 camadas + diario + mensal)
7. Erros padronizados (errorsVerbose)
8. Connection check automatico (Meta)
9. Force refresh analytics

Fase 3: Completo (16-20 semanas)

1. +6 plataformas: Bluesky, Mastodon, Discord, Slack, Google Business, Snapchat
2. Post history import (async)
3. CSV bulk import
4. Reviews import (Google + Facebook)
5. Raw analytics
6. YouTube monetizacao (withBusinessScope)
7. Misc endpoints completos (editar/deletar posts, playlists, mentions)
8. Webhook auto-disable + resend manual

Fase 4: Superior ao bundle.social (20+ semanas)

1. MCP server (model context protocol para AI agents)
2. A2A (Agent-to-Agent) - posts criados por AI
3. Multi-tenant SaaS completo (1 team por cliente)
4. Dashboard com analytics historico (voce armazena, nao 30 dias)
5. Auto-reconnect de contas (detectar + reconectar automaticamente)
6. A/B testing de posts (variacoes de caption/hashtag)
7. AI caption generator (integracao com LLMs)
8. Best time to post (machine learning)
9. Hashtag suggestions
10. Cross-posting inteligente (adaptar formato por plataforma)
11. Approval workflow (rascunho -> revisao -> aprovacao -> schedule)
12. Team collaboration (multi-user com roles)

23. CUSTO DE INFRAESTRUTURA

Custos Mensais Estimados (1000 usuarios)

Servico	Free tier	Custo estimado	Notas
Cloudflare R2	10 GB + 1M ops	~\$5-20	Storage de midia. Zero egress!
PostgreSQL	Neon free 0.5GB	~\$20-50	Neon/Supabase ou self-hosted
Redis	Upstash free 10k cmds/dia	~\$10-30	Rate limits + BullMQ + cache
VPS/Container	Hetzner CX22	~\$5-15	Worker + API server
Cloudflare (CDN)	Free	~\$0	CDN + DNS + WAF
Domain	-.	~\$1	Anual
Email (Resend)	3k/mes free	~\$0-20	Notificacoes
Monitoring (Grafana)	Free	~\$0	Self-hosted
TOTAL	-	~\$40-140/mes	Para 1000 usuarios

Escalabilidade

Custo principal: STORAGE (R2) - cresce com uploads de usuarios.
R2 e MUITO mais barato que S3 (zero egress fee).
Para 10.000 usuarios: ~\$200-500/mes.
Para 100.000 usuarios: ~\$2.000-5.000/mes (principalmente storage).

Worker scaling: BullMQ com Redis permite escalar horizontalmente.
Cada worker processa N jobs em paralelo (concorrenca configuravel).
Adicionar mais workers = mais throughput.

24. DADOS DO SCAN VULNSTRIKE

Endpoints Confirmados pelo Scan

O scan confirmou 114 endpoints da OpenAPI spec. Todos respondem conforme documentado. Nenhuma discrepancia encontrada entre spec e comportamento real.

Headers Observados

Header	Valor	Notas
Content-Type	application/json	Todos endpoints exceto /upload/
x-api-key	pk_live_*	Auth obrigatoria
User-Agent	Variavel	Sem restricao aparente
x-signature	HMAC-SHA256	Apenas em webhooks (outbound)

Observacoes de Seguranca

- 1. API key em header (nao URL) - boa pratica
- 2. HMAC-SHA256 em webhooks - boa pratica
- 3. Rate limits em 3 camadas - boa pratica
- 4. Tokens encriptados no DB - boa pratica
- 5. Sem CORS aberto (verificado pelo scan)
- 6. Sem endpoints nao documentados encontrados
- 7. Swagger JSON publico (intencional, para SDK generation)
- 8. Sem informacoes sensiveis em webhooks (tokens nao vazam)

25. CHECKLIST FINAL DE IMPLEMENTACAO

Auth e Multi-tenancy

- ☐ API Key auth (header x-api-key)
- ☐ Organization > Team > SocialAccount hierarchy
- ☐ Rate limits 3 camadas (Redis)
- ☐ Monthly caps por organization
- ☐ Daily limits por conta real (nao por conexao)

Upload

- ☐ Simple upload (multipart/form-data, 90 MB)
- ☐ Direct upload (presigned URL, 5 GiB, 30min expiry)
- ☐ Multipart upload (64 MiB chunks, 6h expiry, 10k parts max)
- ☐ Upload from URL (1 GB, 60s timeout)
- ☐ Finalize endpoint
- ☐ Multipart sign (re-sign expired parts)
- ☐ Multipart complete (ETags)
- ☐ Multipart abort
- ☐ Auto-cleanup (7 dias deleted, 90 dias unused)

Post

- ☐ POST /post (unificado, multi-plataforma)
- ☐ Validacao por plataforma (aspect ratio, tamanho, duracao)
- ☐ Status: DRAFT -> SCHEDULED -> PROCESSING -> POSTED/ERROR
- ☐ Publisher worker (Promise.allSettled, paralelo por plataforma)
- ☐ Retry transient (3x, backoff exponencial 30s/90s/270s)
- ☐ POST /post/{id}/retry (apos correcao)
- ☐ firstComment automatico (apos publicacao)
- ☐ externalData (IDs e permalinks por plataforma)
- ☐ errorsVerbose (code, userFacingMessage, errorMessage, isTransient)

15 Plataformas

- ☐ Instagram (Meta Graph API, set-channel)
- ☐ Facebook (Meta Graph API, set-channel)
- ☐ TikTok (Content API, OAuth direto)
- ☐ YouTube (Data API v3, set-channel)
- ☐ LinkedIn (Marketing API, set-channel)
- ☐ Twitter/X (API v2, OAuth direto)
- ☐ Threads (Meta, OAuth direto)
- ☐ Bluesky (AT Protocol, serverUrl opcional)
- ☐ Mastodon (API, serverUrl obrigatorio)
- ☐ Pinterest (API v5, sem set-channel)
- ☐ Reddit (API, sem set-channel)
- ☐ Discord (webhook URL)
- ☐ Slack (webhook URL)
- ☐ Google Business (My Business API, set-channel)
- ☐ Snapchat (Marketing API, OAuth direto)

Webhooks

- ☐ 9 eventos (post.published, comment.published, social-account.*, team.*)
- ☐ HMAC-SHA256 signature
- ☐ 15s timeout, 3 tentativas, backoff 30s exponencial
- ☐ 50 concorrentes
- ☐ Auto-disable apos 7 dias sem sucesso
- ☐ Resend manual

Analytics

- ☐ Refresh automatico 24h
- ☐ Force refresh (teams x 5/dia)
- ☐ Raw analytics por plataforma
- ☐ Retencao 30 dias
- ☐ YouTube monetizacao (withBusinessScope)

Comments

- ☐ POST /comment (11 plataformas)
- ☐ Import comentarios (async, 9 plataformas)
- ☐ First comment automatico

Imports

- ☐ Post history import (async, 15 plataformas)
- ☐ CSV bulk import (async)
- ☐ Google reviews import
- ☐ Facebook recommendations import

Misc

- ☐ Editar/deletar posts por plataforma
- ☐ Editar/deletar comentarios por plataforma
- ☐ YouTube playlists + items
- ☐ LinkedIn mentions builder
- ☐ Reddit post-requirements + flairs
- ☐ Google Business location management completo
- ☐ Instagram audio + locations + tags
- ☐ TikTok CML trending list

Cron Jobs

- ☐ Analytics refresh 24h
- ☐ Connection check 6h (Meta)
- ☐ Token refresh proativo
- ☐ Stale upload cleanup diario
- ☐ Analytics purge 30 dias
- ☐ Webhook events purge 7 dias
- ☐ Monthly counter reset dia 1 UTC
- ☐ Webhook auto-disable check
- ☐ Multipart auto-abort 7 dias

CONCLUSAO

Este blueprint documenta COMPLETAMENTE a API bundle.social com todos os valores reais extraídos da OpenAPI spec oficial (114 endpoints) e da documentacao oficial (info.bundle.social). Inclui limites de midia por plataforma (resolucao, duracao, aspect ratio, tamanho), formato de post (campos por plataforma), fluxo completo de postagem (inicio ao fim), paralelismo, rate limits (3 camadas + diario + mensal), webhooks (9 eventos, delivery, retries), erros padronizados, analytics, comments, imports, misc endpoints, schema PostgreSQL completo, stack recomendada, cron jobs, roadmap e checklist final.

Com este documento, um desenvolvedor pode implementar uma alternativa superior ao bundle.social do zero, em outra maquina, sem acessar o site original.